

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 2.3: Area of Part of a Circle — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.3: Area of Part of a Circle
Driving Question	DRIVING QUESTION: How can we use the radius and the central angle of a circular pizza slice to calculate its exact area — and apply the same method to solve real problems in farming, design, and everyday life in Kenya?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Predict — Unequal Slices, Unequal Areas	A circular pizza/chapati was cut into five unequal slices with central angles of approximately 30°, 45°, 60°, 90°, and 135°. The slices were visibly different in size, with the wider-angled slices clearly larger. All slices shared the same radius.	The area of a sector (a slice of a circle) is a fraction of the full circle's area, determined by the ratio of its central angle to 360°. The formula is $A = (\theta/360^\circ) \times \pi r^2$, where θ is the central angle in degrees and r is the radius. A sector with twice the central angle has exactly twice the area — the relationship is directly proportional.	The sector area formula works because a full circle corresponds to a central angle of 360°. Any sector with central angle θ represents the fraction $\theta/360^\circ$ of the full circle. Multiplying this fraction by the full circle area (πr^2) gives the exact area of the sector. This same principle applies to real contexts such as a fan-shaped maize plot in the Rift Valley, a slice of round chapati, or any circular design element.
Lesson 2 OBSERVE & REVIEW — Parts of a Circle, Radius, Central Angle, and the Hunt for a Pattern	Learners measured the radius and central angle of four pre-cut sector cards (all from a circle of radius 10 cm, with angles 45°, 90°, 135° and 180°), calculated the fraction each angle is of 360°, multiplied by the full circle area, and plotted angle against sector area — observing that the data points formed a straight line through the origin.	Learners reviewed and consolidated vocabulary for circle parts: radius, diameter, chord, arc, sector, segment, and central angle. They established that a sector is a fraction ($\theta/360^\circ$) of a full circle and that its area equals that fraction multiplied by πr^2 , producing the expression $A = (\theta/360^\circ) \times \pi r^2$. They also distinguished between minor and major sectors.	The measured data confirms that the area of a pizza slice (sector) grows in exact proportion to its central angle when the radius is fixed — directly addressing the driving question. This proportional relationship, expressed as $A = (\theta/360^\circ) \times \pi r^2$, is the conceptual engine that will be extended to arc length in Lesson 3, formally derived in Lesson 4, and applied to real Kenyan contexts in later lessons.
Lesson 3 EXPLAIN — Arc Length: Deriving the Formula Through the Fraction of	Students measured arc lengths of sectors using string and rulers, then collected class data showing that the ratio of arc length to full circumference always equals theta	Students derived the arc length formula $L = (\theta/360) \times 2\pi r$ from a proportionality statement, applied it to three graded examples including a centre-pivot irrigation	The arc of a sector is a fraction of the full circumference, and that fraction equals the central angle divided by 360 degrees. This same fractional reasoning will govern the

a Circle	divided by 360, confirming that arc length is directly proportional to central angle.	sweep in Nakuru County, and expressed answers to appropriate accuracy.	sector area formula in Lesson 4, because both arc length and sector area represent proportional parts of their respective whole-circle quantities.
Lesson 4 EXPLAIN — Deriving and Applying the Area of a Sector (Minor Sector)	Learners observed that two pizza/chapati slices from the same circular food (radius 14 cm) with central angles 60° and 120° have areas of 102.6 cm ² and 205.2 cm ² respectively — confirming that doubling the central angle exactly doubles the sector area, making the relationship perfectly proportional.	Learners derived and applied the sector area formula $A = (\theta/360^\circ) \times \pi r^2$ by analogy with the arc length formula, understanding that a sector is simply a proportional fraction ($\theta/360^\circ$) of the full circle's area (πr^2).	Learners explained that the formula works because any sector is a fraction of the full circle determined solely by its central angle, and that the same proportional logic that gives arc length also gives sector area — enabling exact calculation of the area of any pizza slice, maize shamba sector, or bead panel wedge given only the radius and central angle.
Lesson 5 EXPLAIN — Area of a Major Sector	A large piece of pizza (or circular cut-out) remaining after a minor sector is removed has a central angle greater than 180 degrees, making it a major sector; its angle and the minor sector angle together always equal 360 degrees.	The formula $A = (\theta \text{ divided by } 360) \times \pi r^2$ calculates the area of any sector — minor or major — because $\theta \text{ divided by } 360$ represents the fraction of the whole circle regardless of the size of the angle; the areas of a major and minor sector of the same circle always sum to πr^2 .	The unwatered portion of a centre-pivot irrigation field in Laikipia and the return-stroke sweep of a matatu windscreen wiper are both real-world major sectors whose areas can be calculated precisely using the same formula derived from the pizza-slicing phenomenon.
Lesson 6 Area of a Minor Segment — The Watermelon Rind Problem	Learners observed a geometric decomposition diagram showing that a minor segment of a circle (modelled as a watermelon rind slice) is formed by removing an isosceles triangle (with two sides equal to the radius r and included angle θ) from a sector of the same circle.	Learners derived and applied the formula: Area of minor segment = $(\theta/360^\circ) \times \pi r^2 - \frac{1}{2}r^2 \sin \theta$, understanding that the sector area gives the 'pie-slice' region and the triangle area ($\frac{1}{2}r^2 \sin \theta$) gives the inner triangular part that must be subtracted to isolate the curved rind region.	Learners explained why the minor segment is always smaller than its corresponding sector (because a positive triangle area is subtracted), applied the formula to Kenyan farming and design contexts (Rift Valley irrigation field and Maasai beadwork brooch), and updated their cumulative circle-area model to include the annotated minor segment with its formula and a real-life Kenyan example.
Lesson 7 Area of a Major Segment — Circular Plot Allocation in the Rift Valley	Learners observed a circular shamba (farm plot) divided by a chord into a small sukuma wiki patch and a large maize field, and worked through three Rift Valley circular plot diagrams (Nakuru, Eldoret, Kericho) to gather evidence about how the major segment relates to the full circle and minor segment.	Learners learned that the area of a major segment = $\pi r^2 - (\text{area of minor segment})$, where area of minor segment = $(\theta/360^\circ) \times \pi r^2 - \frac{1}{2}r^2 \sin \theta$. They applied this five-step process to calculate exact land areas for Rift Valley farming plots and confirmed that the larger region of a circle cut by a chord can be found precisely using only the radius and the central angle.	Learners explained that any circular region divided by a chord produces a minor and a major segment, and that the major segment area is determined by subtracting the minor segment from the total circle area. They connected this to the driving question by demonstrating that — just as with pizza slices — knowing only the radius and central angle is sufficient to calculate the exact area of either segment, enabling precise land allocation, design, and resource planning in real Kenyan contexts.

Lesson 8 Application & Model Building — Solving Real-Life Problems Using Arc Length, Sector Area, and Segment Area	Three real-life Kenyan problems were solved using circle geometry formulas: (1) Mama Wanjiku's circular shamba in Nakuru — a 120° sector of radius 7 m required approximately 206 paving stones (sector area $\approx 51.31 \text{ m}^2$) and an arc-length border of $\approx 14.66 \text{ m}$; (2) Two watermelon slices at Gikomba Market (radii both 20 cm, angles 80° and 160°) — Slice A area $\approx 222.53 \text{ cm}^2$ priced at Ksh 1,112.65 and Slice B area $\approx 445.06 \text{ cm}^2$ priced at Ksh 2,225.30, confirming Slice B costs exactly twice Slice A; (3) A Maasai bead segment in Kajiado ($r = 10 \text{ cm}$, $\theta = 60^\circ$) — sector area $\approx 52.36 \text{ cm}^2$, triangle area $= 43.30 \text{ cm}^2$, segment area $\approx 9.06 \text{ cm}^2$.	All three problems were solved using the same two foundational formulas derived from the chapati phenomenon: Sector Area $= (\theta/360^\circ) \times \pi r^2$ and Segment Area $= \text{Sector Area} - \frac{1}{2}r^2\sin\theta$. The watermelon problem provided direct, numerical confirmation that the sector area formula is LINEAR in θ for a fixed radius — doubling the central angle exactly doubles the area. This validates the proportionality conjecture first raised in Lesson 2 when students noticed that the wider chapati slices appeared to be 'more than just a bit' larger. The segment formula extends the sector formula by subtracting the triangular region formed by the two radii and the chord, isolating only the curved 'bite' between the arc and the chord.	The sector area formula $A = (\theta/360^\circ) \times \pi r^2$ works because a sector is simply a fraction $(\theta/360^\circ)$ of the full circle's area (πr^2). The fraction $\theta/360^\circ$ captures exactly how much of the full rotation the sector occupies — which is why doubling θ exactly doubles the area (for fixed r). The segment formula subtracts the area of the triangle ($\frac{1}{2}r^2\sin\theta$) from the sector area because the triangle occupies the 'filled-in' central portion of the sector that is NOT between the chord and the arc. All three Kenyan problems — the farm, the market, the bead design — are structurally identical to the original chapati phenomenon: a circular region with a known radius is divided by a central angle, and the task is to find the exact area of that portion.
Lesson 9 Digital Tools & Consolidation: Confirming Sector and Segment Areas, Correcting Errors	Students observed that when the same sector and segment calculations from earlier lessons were re-run on scientific calculators and checked against a digital simulation, results either confirmed their hand calculations or revealed specific, diagnosable errors. They also observed four annotated 'Patient Solutions' on the board — flawed worked examples drawn from Kenyan real-world contexts (a Kericho irrigation sector, a Nairobi architectural arc, a Kisumu fishing-net segment) — and saw exactly which line of working introduced each error.	Students learned that: (1) the sector area formula $A = (\theta/360^\circ) \times \pi r^2$ requires θ to be measured in degrees, not radians, and that confusing the two produces a wildly incorrect answer; (2) the segment area requires an explicit subtraction of the triangle area ($A_{\text{segment}} = A_{\text{sector}} - \frac{1}{2}r^2\sin\theta$), and omitting this step gives the sector area, not the segment; (3) using diameter instead of radius in the formula inflates the answer by a factor of 4; (4) premature rounding of π introduces cumulative error in the final answer; (5) digital tools (calculators, simulations) provide independent verification but do not replace the need to understand each step — a calculator will faithfully compute the wrong formula if the wrong formula is entered.	Students explained their understanding by: (1) diagnosing and correcting four flawed worked examples in the Error Autopsy activity, naming the specific error type at the exact line it occurred; (2) producing an Annotated Formula Map linking the sector and segment formulas to labelled diagrams, Kenyan numerical examples, and a 'Common Errors' warning box; (3) updating the Driving Question Board with 'We Now Know' cards that stated generalised rules (not just specific examples) and justifying their answers with formula-based evidence; (4) answering the final cold-call calculation — finding the area of a pizza slice with $r = 28 \text{ cm}$ and $\theta = 135^\circ$ to two decimal places — demonstrating confident, verified application of the formula to the original anchoring phenomenon.
Lesson 10 Assessment and Reflection: Connecting the Pizza Phenomenon to All Learned Concepts	We observed a circular pizza cut into unequal slices in Lesson 1 and noticed that wider-angled slices appeared larger. Today we observed a circular shamba in Nakuru divided into a fenced sector of 80 degrees and an unfenced major sector, and we measured (calculated) the arc length, sector areas, and segment areas using radius 21 m and central angle 80 degrees.	We learned that arc length $= (\theta/360) \times 2\pi r$; sector area $= (\theta/360) \times \pi r^2$; major sector area $= \text{total circle area} - \text{minor sector area}$; segment area $= \text{sector area} - \frac{1}{2}r^2\sin\theta$; and major segment area $= \text{total circle area} - \text{minor segment area}$. We also learned that these formulas apply directly to real Kenyan contexts including pizza	We explained why a pizza slice with twice the angle is exactly twice the area because both quantities are directly proportional to the central angle as a fraction of 360 degrees. We explained why the segment formula requires subtracting the triangle area and how omitting that step gives the sector area by mistake. We explained how every formula we derived traces back to the

		pricing, shamba allocation, maize planting density, and circular irrigation field management.	original question posed in Lesson 1: can we predict the exact area of any slice if we know only the radius and the angle of the cut?
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ATTRIBUTION AND LICENSING

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